

Accelerating Low-Frequency Convergence for Limited-Angle DBT

Two-Channel Fidelity in PDHG

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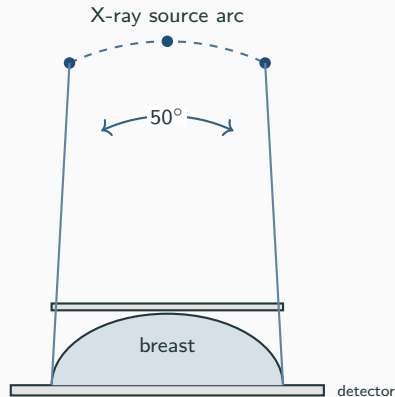
Motivation

What is limited-angle reconstruction?

- Tomography recovers a cross-section from X-ray projections taken at many source angles around the object.
- **Sufficient sampling.** For a fan-beam scan, a source arc of 180° plus the fan angle already measures every line integral; a full 360° rotation only re-measures each ray from the opposite side.
- **Limited-angle reconstruction (LAR)** uses a source arc *below* that threshold. The measurements no longer determine the image: a wedge of spatial frequencies is never observed.
- Digital breast tomosynthesis (DBT) operates deep in this regime, with a source arc of only $[15^\circ, 50^\circ]$, so its reconstruction is severely ill-posed.

Why limited-angle?

- **Dose.** Fewer views means less ionizing radiation per scan.
- **Geometry.** In DBT the detector stays under the compressed breast while the X-ray tube sweeps a short arc; a full gantry rotation is mechanically impossible.
- **Acquisition time.** A short sweep is fast, limiting motion blur and patient discomfort.
- **Use cases.** DBT for early breast-cancer detection; also lung-nodule and kidney-stone imaging.



Two reconstruction settings: 2D and 3D

The two-channel method is evaluated in both geometries:

- **2D fan-beam.** A single slice; cheap enough to sweep image resolution and noise level.
- **3D cone-beam.** The actual DBT geometry; full-volume reconstruction on anatomical phantoms.

Both are severely under-determined: the number of unknowns far exceeds the number of measurements, increasingly so at finer resolution.

Slice resolution	Unknowns (pixels)	Measurements (rays)
512^2	262,144	25,600
256^2	65,536	25,600
128^2	16,384	25,600

Measurements = 25 views $\times$ 1024 detector bins.

The reconstruction setup

Reconstruction with PDHG (Chambolle–Pock)

Minimize directional-TV and ℓ_1 sparsity, subject to the projections matching the data within a tolerance, with $f \geq 0$.

- The objective is non-differentiable (ℓ_1 , TV) and hard-constrained: gradient descent does not apply.
- PDHG alternates a *dual* step (pull the data residual toward the tolerance ball) and a *primal* step (pull the image toward sparsity and $f \geq 0$). Cheap projections and proximal maps, no derivatives.

Step sizes τ and σ are coupled through the operator norm $\|K\|$; see appendix.

Single-channel data fidelity

Prior work¹ solves a constrained DTV / ℓ_1 problem with a square-root Hanning filter on the sinogram residual:

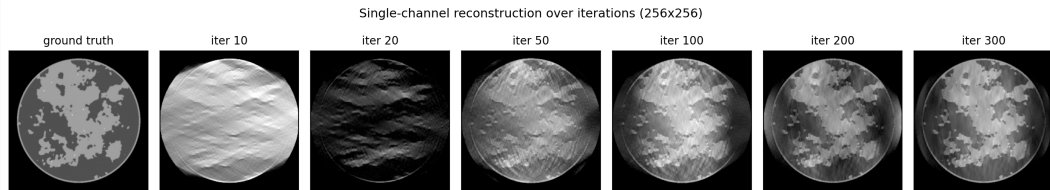
$$\begin{aligned} \min_{f \geq 0} \quad & \alpha_x \|\partial_x f\|_1 + \alpha_y \|\partial_y f\|_1 + \alpha_z \|\partial_z f\|_1 + \beta \|f\|_1 \\ \text{s.t.} \quad & \|R_{[c]}(g - Xf)\|_2 \leq \varepsilon \sqrt{|g|}. \end{aligned}$$

- $R_{[c]} = \text{Hann}^{1/2}(c)$ attenuates low spatial frequencies in the residual (noise suppression by design).
- Solved with PDHG and He–Yuan over-relaxation ($\rho \approx 1.75$).

¹E. Y. Sidky et al., *J. Med. Imaging* 12(S1):S13013, 2025.

The problem

What happens during a single-channel run



- Fine structure (calcifications, glandular speckling) settles fast.
- Large soft-tissue regions stay blurred, with a slowly-varying intensity error, through iter ~ 100 –300.
- This is the low-frequency convergence bottleneck.

2D fan-beam · 25 views · 50° arc · 256² grid

Why does this happen?

In the detector-frequency domain, the data-fidelity weight is

$$W(g - Xf), \quad W = R_{[c]} = \text{Hann}^{1/2}(c).$$

- PDHG corrects mode k at a rate $\propto s_k^2$ (squared singular values of WX).
- In LAR DBT, s_k is already tiny for low k (ill-conditioning).
- $\text{Hann}^{1/2}$ shrinks low- k further *by design*.
- Stability bound is dictated by the *largest* s_k (high frequencies).

Consequence

The optimizer is forced to use step sizes that under-drive the very modes that are hardest to recover.

The idea: two-channel fidelity

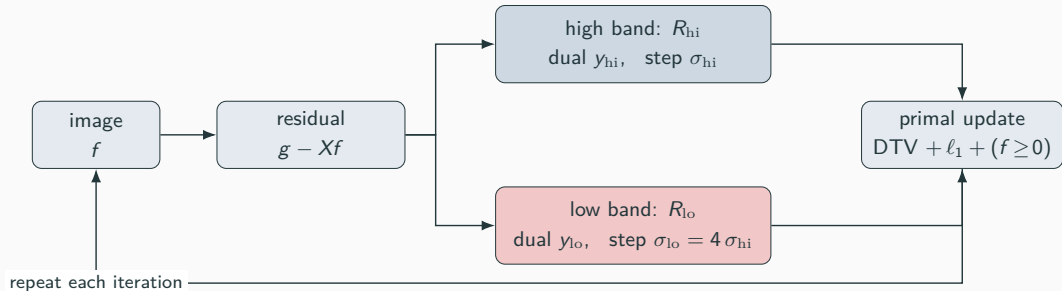
Split the sinogram into two bands

$$\begin{aligned} & \min_{f \geq 0} \alpha_{\bullet} \|\partial_{\bullet} f\|_1 + \beta \|f\|_1 \\ \text{s.t.} \quad & \begin{cases} \|R_{\text{hi}}(g - Xf)\|_2 \leq \varepsilon_{\text{hi}} \sqrt{|g|}, \\ \|R_{\text{lo}}(g - Xf)\|_2 \leq \varepsilon_{\text{lo}} \sqrt{|g|}. \end{cases} \end{aligned}$$

- $R_{\text{hi}} = \text{Hann}^{1/2}(c)$ – preserves the high-frequency channel from prior work.
- $R_{\text{lo}} = \text{Hann}^{1/2}(c_{\text{lo}})$ with $c_{\text{lo}} \gg c$ – a narrow-band low-pass channel that targets the slow modes.
- Independent dual variables $y_{\text{hi}}, y_{\text{lo}}$ with independent step sizes $\sigma_{\text{hi}}, \sigma_{\text{lo}}$.

Step sizes follow the same balance heuristic as the single-channel paper: $\tau \sigma_{\text{hi}} \|K\|^2 \approx 1$, with $\sigma_{\text{lo}} = 4\sigma_{\text{hi}}$ to drive the slow modes harder.

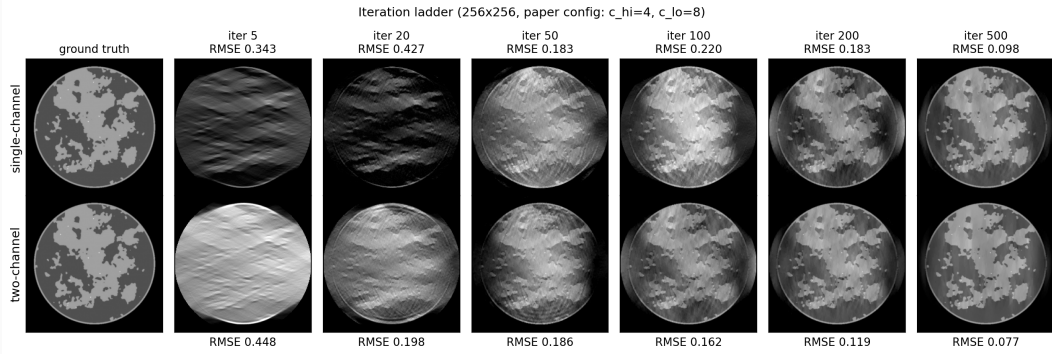
The two-channel iteration



- The data residual is split into a high- and a low-frequency band.
- Each band carries its own dual variable and its own step size.
- The low band is driven $4\times$ harder ($\sigma_{lo} = 4 \sigma_{hi}$), so the slow modes receive the larger correction.
- The primal update and the image itself are unchanged from single-channel.

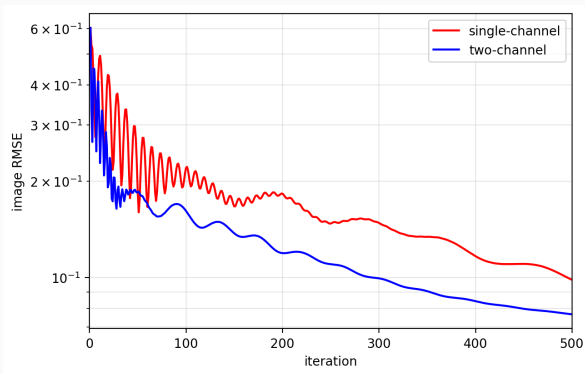
Results

2D fan-beam: single vs two-channel



- Two-channel resolves recognizable soft-tissue structure several iterations earlier than single-channel.

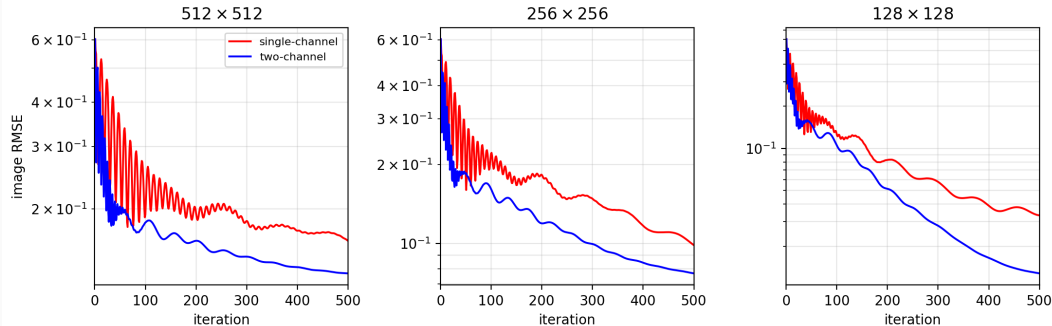
Convergence behaviour



- Single-channel oscillates strongly through iter ~ 200 ; two-channel converges smoothly from the start.

2D fan-beam · 25 views · 50° arc · 256^2 grid

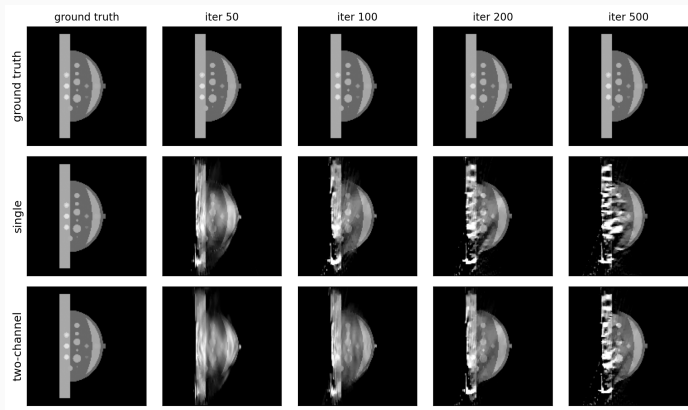
Persistence across image sizes



- Two-channel sits below single-channel at every resolution.
- Iter-500 image-RMSE reduction: 19%, 22%, 62% at 512^2 , 256^2 , 128^2 .

2D fan-beam · 25 views · 50° arc · 512^2 , 256^2 , 128^2 grids

3D analytic breast: iteration ladder



Mid-axial slice across iterations. Two-channel resolves the discrete fatty-mass inserts earlier; image-RMSE reduction reaches 35% at iter 200 before semi-convergence sets in.

3D cone-beam · 25 views · 50° arc · 144 × 144 × 32 grid

3D analytic breast: convergence

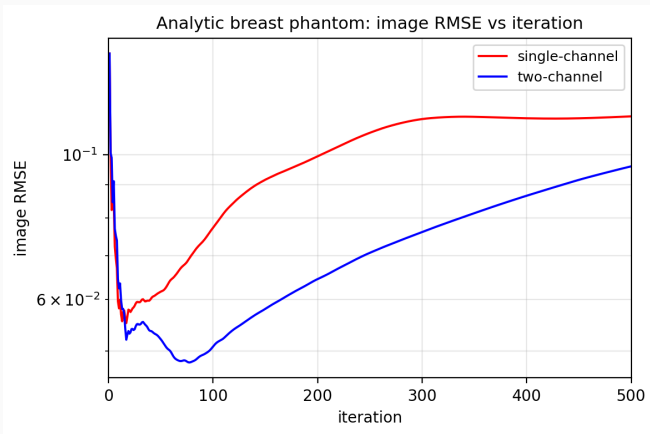
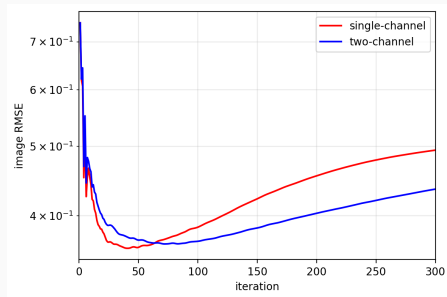
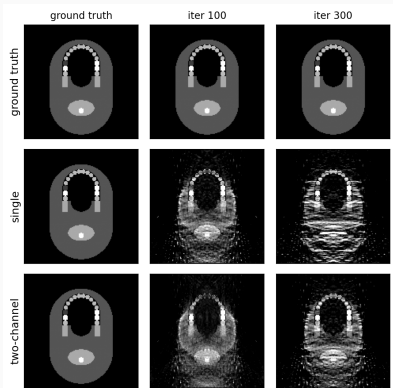


Image RMSE vs. iteration. Two-channel leads throughout: 34% reduction at iter 100, peaking near 35% at iter 200.

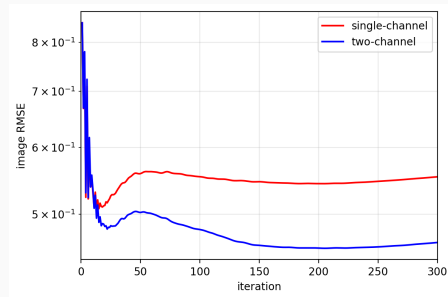
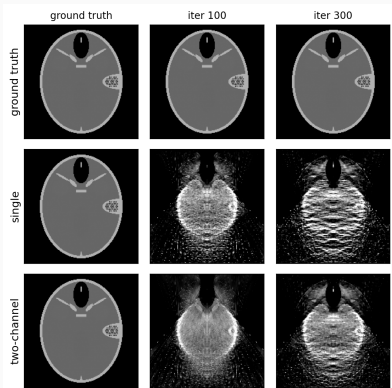
Analytic jaw phantom



Two-channel resolves the teeth and gold-crown contours earlier; iter-300 image-RMSE reduction is 12% (single 0.494, two-channel 0.436).

3D cone-beam · 25 views · 100° arc · 144 × 144 × 80 grid

Analytic head phantom



Two-channel stays below single-channel throughout; iter-300 image-RMSE reduction is 16% (single 0.554, two-channel 0.463).

3D cone-beam · 25 views · 100° arc · 176 × 176 × 16 grid

Conclusion

- LAR reconstruction has a *mode-dependent* convergence bottleneck: low spatial frequencies are starved of corrective drive because the PDHG step size is bound by high-frequency stability.
- Splitting the data fidelity into a low-pass and a high-pass channel, each with its own dual variable and step size, accelerates low-frequency settling without breaking stability.
- Confirmed at a clinical 50° arc on the 2D breast phantom (iter-100 image-RMSE 26% below single-channel) and the 3D analytic breast (34% below). The gain persists at wider arcs on the head and jaw phantoms.

- **Dyadic multi-channel fidelity.** Replace the two bands with a partition-of-unity bank of dyadic band-pass filters whose supports halve toward DC, with a matched geometric step schedule $\sigma_i = 2^i \sigma_0$. The product-space operator norm grows linearly (not exponentially) in the number of channels. Manuscript in preparation.
- **Theorem-compliant step sizes.** Derive the per-band step sizes from a weighted power iteration so the empirical balance heuristic connects to the product-space PDHG convergence guarantee.
- **Tuning the low-frequency constraint.** The per-band data-tolerance ratio that minimises image RMSE is not the same on 2D and 3D phantoms. Understanding what problem features control it is open.

Questions?

Backup: per-band tolerance tuning

The per-band tolerance ratio

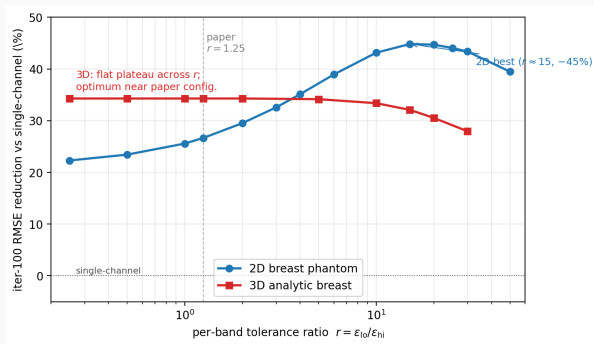
Each channel constrains its band of the data residual to lie inside an ℓ_2 ball,

$$\|R_{\text{hi}}(g - Xf)\|_2 \leq \varepsilon_{\text{hi}} \sqrt{|g|}, \quad \|R_{\text{lo}}(g - Xf)\|_2 \leq \varepsilon_{\text{lo}} \sqrt{|g|}.$$

The ratio $r = \varepsilon_{\text{lo}}/\varepsilon_{\text{hi}}$ sets how tightly the low-frequency residual is held relative to the high-frequency residual:

- Small r tightens the LF data fit; the recon is pulled closer to the data in the low-frequency band.
- Large r loosens the LF constraint; the LF prior (DTV, ℓ_1 , non-negativity) carries more weight.
- Changing r moves the *fixed point* of the optimization, not just the trajectory. The original paper used $r = 1.25$; the next slide sweeps it to see how much performance is available.

Tolerance-ratio sweep across phantoms



- 2D breast: clear interior optimum near $r \approx 15$ (-45% vs single-channel); paper $r=1.25$ leaves $\sim 20\%$.
- 3D analytic breast: flat plateau at -34% ; paper config is already near the optimum.
- The optimum does not transfer between dimensions. Band-cutoff is empirically robust on both (sweep omitted).

References

- E. Y. Sidky et al., *Accurate volume image reconstruction for DBT with directional-gradient and pixel sparsity regularization*, J. Med. Imaging 12(S1):S13013, 2025.
- A. Chambolle, T. Pock, *A first-order primal-dual algorithm for convex problems with applications to imaging*, J. Math. Imag. Vision 40(1):120–145, 2011.
- B. He, X. Yuan, *Convergence analysis of primal-dual algorithms for a saddle-point problem: from contraction perspective*, SIAM J. Imag. Sci. 5(1):119–149, 2012.
- Z. Zhang, B. Chen, D. Xia, E. Y. Sidky, X. Pan, *Directional-TV algorithm for image reconstruction from limited-angular-range data*, Med. Image Anal. 70:102030, 2021.